AssemblyAI Voice Agent Hackathon 2026

MIRROR

See what a voice command will do before it does it.

A counterfactual runtime for voice agents, built on AssemblyAI’s Voice Agent API and LLM Gateway.

Voice agents jump from language to action

“Move 700 XR-17 from Mumbai to the Pune site.”

There are two Pune sites. A typical agent guesses, or asks a generic “which one?”, and then writes to the inventory system. If it guessed wrong, nobody finds out until the stock is in the wrong place.

Guess

Picks a site and moves real stock on a coin flip.

Confirm everything

“Are you sure?” on every step, until people say yes without reading.

Trust the reply

“Done!” is the system’s claim, not a check that reality agrees.

One sentence, two simulated futures

“Move 700 XR-17 from Mumbai to the Pune site.”

If you meant Pune Site 1

1200 to 1900

Blocked: 400 over the site’s capacity of 1500

If you meant Pune Site 2

300 to 1000

Valid: Mumbai stays above safety stock

Nothing has moved. The agent asks the question whose answer changes the outcome: “Site 1 would exceed capacity by 400 units; Site 2 remains valid. Which one do you mean?”

Explore freely, commit exactly, verify always

1

Fork

Every reading the ERP’s own data allows. Unknown names resolve to nothing, never a guess.

2

Simulate

Against a frozen snapshot. Zero writes, deterministic, under a millisecond per future.

3

Clarify

Only the question that changes the outcome, with the reason in numbers.

4

Explore

“What if 900?”, “compare 500 and 900”, “as much as is safe”. Reality stays untouched.

5

Commit

Only on the user’s own “commit”, with a token bound to one version, its evidence and its simulation.

6

Verify

Read the whole world back. Anything but an exact match is “diverged”, with a rollback ready.

When reality disagrees, MIRROR says so

Reality diverged

The ERP reported success but short-shipped: it moved 450, not 500.

Cell	Expected	Actual
Mumbai	1200	1250
Pune Site 2	800	750

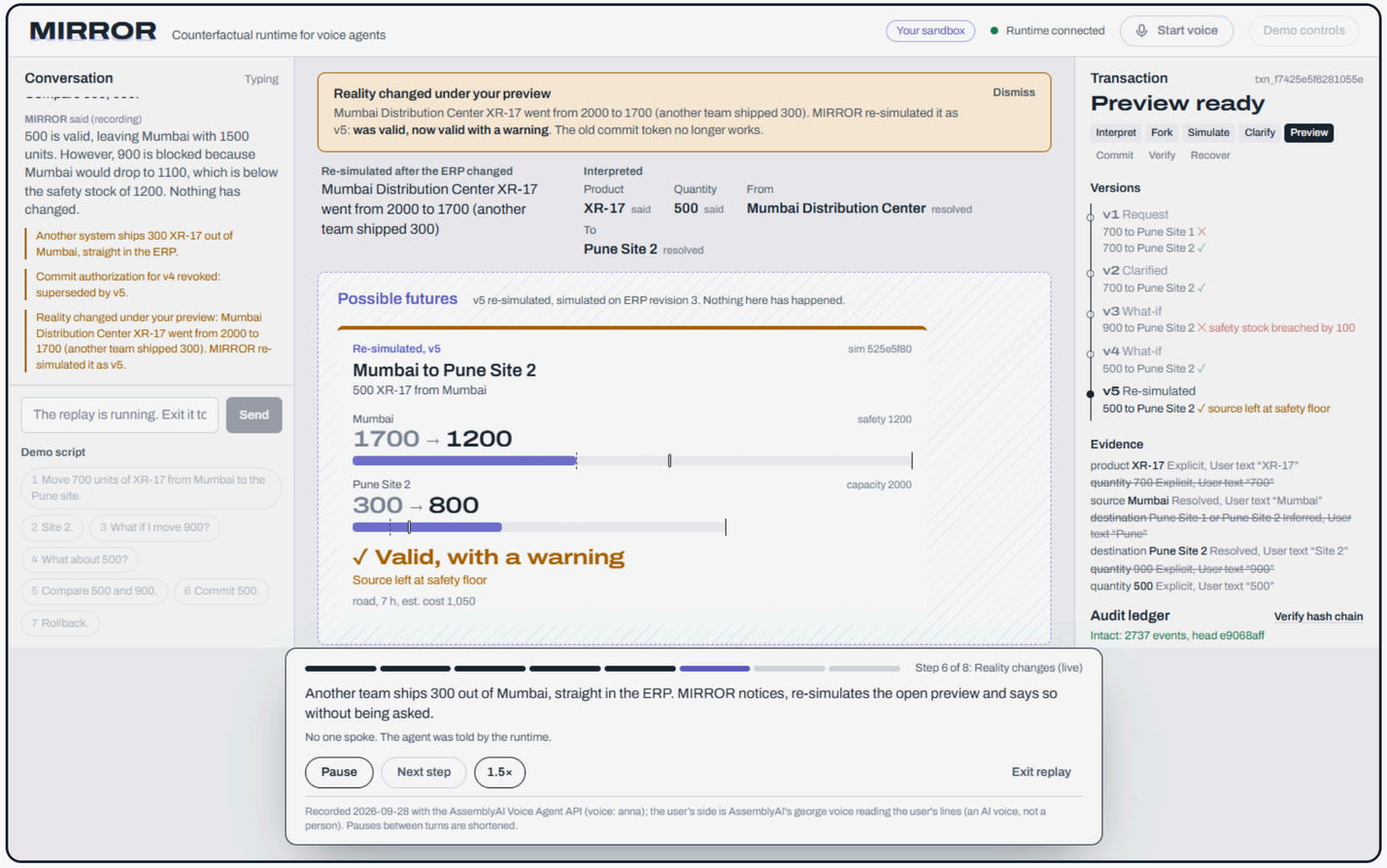
Never marked verified. The rollback is computed from what actually happened, then verified again.

Reality changed under you

Another team shipped 300 out of Mumbai while a preview was open. MIRROR re-simulated it, revoked the old commit token, and the agent spoke up without being asked:

> *“Heads up: reality changed under your preview. Mumbai XR-17 went from 2000 to 1700. I re-simulated it.”*

Prediction above the line, reality below



Hatched means simulated

Futures, what-ifs and comparisons. Nothing in that area has happened.

Solid means the ERP

Read back after every write, with the reality status on the line itself.

Every step on record

Immutable versions and a SHA-256 hash-chained audit ledger.

How we use AssemblyAI

Voice Agent API

- Recognition, turn detection, barge-in, tool calling and speech over one WebSocket, relayed by our server so the key never reaches the browser
- Every tool.call is a proposal: held until reply.done, dropped if the user interrupts
- Tools follow state: commit exists only while a valid preview does
- Keyterms narrow to the options being asked about
- reply.create lets the agent announce reality changes unprompted

LLM Gateway

- Reads typed phrasing the rule-based parser can’t
- Its JSON goes through the same tools and the same evidence gate
- Live: “yep sounds right to me” came back as commit. The backend refused.

The model proposes. The backend decides.

If the model tries to…	MIRROR
pick Site 1 when you said “the Pune site”	keeps both futures simulated and asks you
invent a quantity you never said	marks it inferred; it needs your confirmation
commit because you said “sounds good”	refuses: commit needs your own word
commit a version you did not see	rejects the token, which is bound to one version
run a tool call you talked over	drops it before it reaches the runtime
say “done” when it is not	shows the ERP as read back after the write

Measured, not claimed

116
backend tests, including adversarial ones

53/53
live AssemblyAI checks, 39 spoken turns

28/28
live, with the spoken reality alert

13/13
real browser, microphone path, live

26 scenarios	Naive voice agent	MIRROR
Correct final state	10	26
Unsafe actions taken	9	0
Wrong interpretations executed	6	0

Why it matters

Operators

Voice where hands are busy: warehouses, field service, dispatch. Without a misheard number moving real stock.

The business

Wrong transfers caught before they happen, every decision traceable, and a way to give voice agents write access safely.

Beyond inventory

Action, state, simulator, consequences, commit, verifier. The same loop fits procurement, access changes, cloud operations and scheduling.

A human in the loop by consequence, not by confirmation fatigue.

Agents are getting write access. Fast.

AI agents

$7.8B to $52.6B

2025 to 2030, 46.3% a year

Warehouse management systems

$4.6B to $10.0B

2025 to 2030, 17.1% a year

Beachhead: inventory and warehouse teams adding voice to their WMS or ERP, where hands are busy and a wrong transfer is expensive. **Then:** every agent allowed to change a system of record.

Sources: MarketsandMarkets, AI Agents Market report and Warehouse Management System Market report, 2025 to 2030 forecasts.

How MIRROR makes money

Seats

A monthly subscription per operator using the voice console, including the audit ledger.

Verified transactions

Usage pricing per committed and verified change, so cost follows the value delivered.

Connectors

Paid adapters for the systems customers already run, such as SAP, NetSuite and major WMS suites.

Private deployment

An annual licence to run it on premises, for regulated and security-sensitive operations.

Go to market: mid-size distributors first, through the integrators who already install their WMS and ERP.

Where today's options stop

Approach	What it does well	What it misses
Voice agents with tool calling	Natural conversation, fast action	Acts on the model's reading
"Are you sure?" confirmations	A human in the loop	No consequences shown; people click yes
Prompt and output guardrails	Blocks unsafe text	Never sees the effect on real data
Workflow automation (RPA)	Reliable scripted steps	No ambiguity, no what-ifs
MIRROR	**Simulates every reading, asks only what matters**	**Commits exactly one version, then verifies reality**

What comes next

Next 3 months

- NetSuite and SAP Business One connectors
- Approval routing for high-risk futures
- Hindi and other languages AssemblyAI supports

6 to 12 months

- Multi-step changes: batch transfers, purchase orders
- A policy editor for operations managers
- SOC 2 and private deployment

Beyond inventory

- Procurement
- Access and permission changes
- Cloud operations and scheduling

It scales cheaply: a simulation takes under a millisecond, and only the ERP adapter is domain-specific.

Try it

prajwalsutar899-mirror.hf.space

Watch

A 2½-minute replay of a live AssemblyAI session, recomputed by the server as you watch

Talk

Your own voice, in the browser, against a private sandbox copy of the ERP

Break it

Demo controls arm a short shipment or change the world under your preview

Code, tests and live results: the GitHub link in this submission